KAIZEN MATHS

Lesson starter

Surds and Radicals

Surd Operations

Name _____

S1

Simplify: $4\sqrt{5} \times 4\sqrt{2}$

S2

Simplify: $\frac{5\sqrt{15}}{4\sqrt{3}}$

S3

Simplify: $6\sqrt{2} + 6\sqrt{2}$

S4

Simplify: $4\sqrt{27} - 2\sqrt{12} + 2\sqrt{28} + 2\sqrt{7}$

Show your working

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ALG-A602-L2 · S

Show your working

KAIZEN MATHS

Exit ticket

Surds and Radicals

Surd Operations

Name _____

E1

Simplify: $4\sqrt{3} \times 3\sqrt{2}$

E2

Simplify: $\frac{5\sqrt{25}}{5\sqrt{5}}$

E3

Simplify: $4\sqrt{11} + 2\sqrt{11}$

KAIZEN MATHS

Exit ticket

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ALG-A602-L2 · E

Show your working

KAIZEN MATHS

Exit ticket

Surds and Radicals

Surd Operations

Name _____

E1

Simplify: $4\sqrt{3} \times 3\sqrt{2}$

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Simplify: $\frac{5\sqrt{25}}{5\sqrt{5}}$

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Simplify: $4\sqrt{11} + 2\sqrt{11}$

ALG-A602-L2 · E

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Show your working

Teacher answers & working

Surds and Radicals · Surd Operations

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S1

Simplify: $4\sqrt{5} \times 4\sqrt{2}$

Answer:

$16\sqrt{10}$

Working:

Multiply coefficients and surd parts separately

$4 \times 4 = 16$

$\sqrt{5} \times \sqrt{2} = \sqrt{(5 \times 2)} = \sqrt{10}$

Combine

$16\sqrt{10}$

Simplify the resulting surd

$\sqrt{10} = \sqrt{10}$

$16\sqrt{10}$

S2

Simplify: $\frac{5\sqrt{15}}{4\sqrt{3}}$

Answer:

$\frac{5\sqrt{5}}{4}$

Working:

Separate coefficients and surds

$\frac{5}{4} \times \frac{\sqrt{15}}{\sqrt{3}}$

Simplify the coefficient fraction

$\frac{5}{4} = \frac{5}{4}$

Divide the surds

$\frac{\sqrt{15}}{\sqrt{3}} = \sqrt{\frac{15}{3}} = \sqrt{5}$

Simplify if needed

$\sqrt{5} = \sqrt{5}$

$\frac{5\sqrt{5}}{4}$

Teacher answers & working

Surds and Radicals · Surd Operations

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S3

Simplify: $6\sqrt{2} + 6\sqrt{2}$

Answer:

$12\sqrt{2}$

Working:

Identify like surds

Both terms contain $\sqrt{2}$

Combine the coefficients

$6 + 6 = 12$

Keep the surd part unchanged

$12\sqrt{2}$

S4

Simplify: $4\sqrt{27} - 2\sqrt{12} + 2\sqrt{28} + 2\sqrt{7}$

Answer:

$8\sqrt{3} + 6\sqrt{7}$

Working:

Simplify each surd term

$4\sqrt{27} = 12\sqrt{3}$
$-2\sqrt{12} = -4\sqrt{3}$
$2\sqrt{28} = 4\sqrt{7}$
$2\sqrt{7} = 2\sqrt{7}$

Group like surds

Terms with the same $\sqrt{}$ part can be combined.

Combine coefficients

$8\sqrt{3} + 6\sqrt{7}$

Teacher answers & working

Surds and Radicals · Surd Operations

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E1

Simplify: $4\sqrt{3} \times 3\sqrt{2}$

Answer:

$12\sqrt{6}$

Working:

Multiply coefficients and surd parts separately

$4 \times 3 = 12$

$\sqrt{3} \times \sqrt{2} = \sqrt{(3 \times 2)} = \sqrt{6}$

Combine

$12\sqrt{6}$

Simplify the resulting surd

$\sqrt{6} = \sqrt{6}$

$12\sqrt{6}$

E2

Simplify: $\frac{5\sqrt{25}}{5\sqrt{5}}$

Answer:

$\sqrt{5}$

Working:

Separate coefficients and surds

$\frac{5}{5} \times \frac{\sqrt{25}}{\sqrt{5}}$

Simplify the coefficient fraction

$\frac{5}{5} = 1$

Divide the surds

$\frac{\sqrt{25}}{\sqrt{5}} = \sqrt{\frac{25}{5}} = \sqrt{5}$

Simplify if needed

$\sqrt{5} = \sqrt{5}$

$\sqrt{5}$

Teacher answers & working

Surds and Radicals · Surd Operations

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E3

Simplify: $4\sqrt{11} + 2\sqrt{11}$

Answer:

$6\sqrt{11}$

Working:

Identify like surds

Both terms contain $\sqrt{11}$

Combine the coefficients

$4 + 2 = 6$

Keep the surd part unchanged

$6\sqrt{11}$